10. An inductor of inductance L carries an electric current i. How much energy is stored while establishing the current in it?
11. Show that the mutual inductance between a pair of coils is same ($M_{12} = M_{21}$).
12. How will you induce an emf by changing the area enclosed by the coil?
13. Show mathematically that the rotation of a coil in a magnetic field over one rotation induces an alternating emf of one cycle.
14. Elaborate the standard construction details of AC generator.
15. Explain the working of a single-phase AC generator with necessary diagram.
16. How are the three different emfs generated in a three-phase AC generator? Show the graphical representation of these three emfs.
17. Explain the construction and working of transformer.
18. Mention the various energy losses in a transformer.
19. Give the advantage of AC in long distance power transmission with an illustration.
20. Find out the phase relationship between voltage and current in a pure inductive circuit.
21. Derive an expression for phase angle between the applied voltage and current in a series RLC circuit.
22. Define inductive and capacitive reactance. Give their units.
23. Obtain an expression for average power of AC over a cycle. Discuss its special cases.
24. Explain the generation of LC oscillations in a circuit containing an inductor of inductance L and a capacitor of capacitance C.
25. Prove that the total energy is conserved during LC oscillations.
26. Compare the electromagnetic oscillations of LC circuit with the mechanical oscillations of block-spring system qualitatively to find the expression for angular frequency of LC oscillator.

IV. Numerical problems

1. A square coil of side 30 cm with 500 turns is kept in a uniform magnetic field of 0.4 T. The plane of the coil is inclined at an angle of 30° to the field. Calculate the magnetic flux through the coil. (Ans: 9 Wb)
2. A straight metal wire crosses a magnetic field of flux 4 mWb in a time 0.4 s. Find the magnitude of the emf induced in the wire. (Ans: 10 mV)
3. The magnetic flux passing through a coil perpendicular to its plane is a function of time and is given by $\Phi_B = (2t^3 + 4t^2 + 8t + 8)$ Wb. If the resistance of the coil is 5 Ω, determine the induced current through the coil at a time t = 3 second. (Ans: 17.2 A)
4. A closely wound circular coil of radius 0.02 m is placed perpendicular to the magnetic field. When the magnetic field is changed from 8000 T to 2000 T in 6 s, an emf of 44 V is induced in it. Calculate the number of turns in the coil. (Ans: 35 turns)

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